

Scalability & Future Plan

Date	Team ID	Project Name	Maximum Marks
		SB Stocks	1

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	Alpha Vantage free tier API rate limit	Limits real-time data for multiple concurrent users	High
2	SQLite single-file database	Not suitable for high-concurrency production use	High
3	No real-time push updates	Users must refresh for latest prices	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade/Solution
1	User Load	Supports ~50 concurrent users	Deploy with load balancer, horizontal scaling
2	Data Storage	SQLite local database	Migrate to PostgreSQL with connection pooling
3	Performance	Synchronous API calls	Implement Redis caching and async task processing
4	Security	Basic session auth	Add JWT tokens, rate limiting, HTTPS enforcement

Future Roadmap

Phase	Planned Feature/Enhancement	Target Timeline	Expected Impact
Phase 1	AI-powered stock recommendations	Q3 2026	Personalized trading suggestions
Phase 2	Real-time WebSocket price feeds	Q4 2026	Live price updates without refresh
Phase 3	Mobile app (React Native)	Q1 2027	Wider user accessibility
Phase 4	Social trading and leaderboards	Q2 2027	Community engagement and gamification